MSG No. 86.70 $P4_2/n'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,4,6,7]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[2,3,5,8]`

Table 2: Wyckoff site: `2b`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,4,6,7]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,5,8]`

Table 3: Wyckoff site: `4c`, site symmetry: `-1'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,6]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,7]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,8]`

Table 4: Wyckoff site: `4d`, site symmetry: `-1'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,5]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,6]`
3	$\left[\frac{1}{2}, 0, 0\right]$	`[3,7]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,8]`

Table 5: Wyckoff site: `4e`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,4]`
2	$\left[\frac{3}{4}, \frac{1}{4}, z + \frac{1}{2}\right]$	`[2,3]`
3	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[5,8]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2} - z\right]$	`[6,7]`

Table 6: Wyckoff site: `4f`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,4]`
2	$\left[\frac{3}{4}, \frac{3}{4}, z + \frac{1}{2}\right]$	`[2,3]`
3	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[5,8]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - z\right]$	[6,7]

Table 7: Wyckoff site: 8g, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$\left[-y, x + \frac{1}{2}, z + \frac{1}{2}\right]$	[2]
3	$\left[y + \frac{1}{2}, -x, z + \frac{1}{2}\right]$	[3]
4	$\left[\frac{1}{2} - x, \frac{1}{2} - y, z\right]$	[4]
5	$[-x, -y, -z]$	[5]
6	$\left[y, \frac{1}{2} - x, \frac{1}{2} - z\right]$	[6]
7	$\left[\frac{1}{2} - y, x, \frac{1}{2} - z\right]$	[7]
8	$\left[x + \frac{1}{2}, y + \frac{1}{2}, -z\right]$	[8]